Load Continental Trait Records from VegVault

Description

Loads raw trait data from VegVault for a single continental bounding box, removes incomplete records, and translates numeric taxon IDs to taxon names. Optionally prints progress messages.

Usage

```
load_continental_trait_records_from_vegvault(
  data_continental_unit,
  vec_trait_domain_names,
  path_vegvault = here::here("Data/Input/VegVault.sqlite"),
  verbose = TRUE
)
```

Arguments

`data_continental_unit`	A single-row data frame (one continental unit from the spatial grid) with columns 'scale_id', 'x_min', 'x_max', 'y_min', and 'y_max'.
`vec_trait_domain_names`	A non-empty character vector of trait domain names to extract (e.g. 'c("Specific leaf area (SLA)", "Plant height vegetative")').
`path_vegvault`	A single character string giving the path to the VegVault SQLite database (default: 'here::here("Data/Input/VegVault.sqlite")').
`verbose`	A single logical value. If 'TRUE' (default), progress messages are printed via 'cli::cli_inform()'. Set to 'FALSE' to suppress all console output.

Details

The function performs the following steps:

1. Validates all input arguments. 2. Derives the bounding-box limits and scale identifier from 'data_continental_unit'. 3. Calls [load_trait_records_from_vegvault()] to retrieve raw records. 4. Calls [filter_complete_trait_records()] to drop incomplete rows. 5. Calls [resolve_trait_taxon_ids()] to replace numeric IDs with taxon names. 6. Returns the final tibble with resolved names and source IDs.

Value

A tibble containing 'taxon_name', source identifiers ('dataset_id', 'dataset_name', 'data_source_id', 'sample_id', 'trait_id', and 'taxon_id'), and the cleaned trait fields for the supplied continental unit.

See Also

[load_trait_records_from_vegvault()], [filter_complete_trait_records()], [resolve_trait_taxon_ids()]